Project Proposal: Flexibility Potential of a Swiss MV Distribution Grid Under Future DER Scenarios

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Article On the Flexibility Potential of a Swiss Distribution Grid: Opportunities and Limitations [1]

Case Study The case study focuses on a synthetic medium-voltage (MV) distribution grid located in the area of Eschenbach SG in the Canton of St. Gallen, Switzerland. The grid is identified as `20_0` in the Swiss Power Distribution Grids (Swiss-PDGs) database [3], a publicly available, geo-referenced dataset of 879 synthetic MV grids covering all of Switzerland, generated from open geodata and validated against real reference grids.

The network operates at 20 kV and is supplied by a 25 MVA HV/MV transformer. It comprises 51 buses, 50 lines (38 underground cables and 12 overhead lines), and 26 load nodes representing MV/LV transformer substations. Under peak simultaneous loading conditions, the total active demand reaches 7.60 MW, with a maximum line loading of 97.2% and six lines loaded above 60%, indicating that network constraints are binding in stressed conditions. The grid exhibits a typical radial topology with three distinct feeders departing from a central Point of Common Coupling (PCC), making it structurally analogous to the Walenstadt grid studied in the reference article.

Distributed energy resources (DERs) — rooftop photovoltaic (PV) systems, battery energy storage systems (BESS), and heat pumps (HP) — are assigned to the grid nodes using the companion open dataset *Future Deployment and Flexibility of Distributed Energy Resources in the Distribution Grids of Switzerland* [4], which provides node-level DER allocations and hourly-resolved profiles for three reference years (2030, 2040, and 2050), consistent with the Swiss Energieperspektiven 2050+ targets. Non-controllable load profiles and outdoor temperature time series from Swiss weather stations are also taken from this dataset. All data sources are openly available under CC-BY 4.0.

Research Question The growing deployment of distributed renewable generation and flexible loads increases the flexibility potential of distribution grids, yet this potential is constrained by network topology and physical limits. We investigate how grid constraints shape the aggregated flexibility of a Swiss MV distribution grid, and how this flexibility evolves as DER penetration increases from 2030 to 2050.

Unlike existing studies that rely on artificial benchmark topologies, we use a geo-referenced synthetic grid populated with DER projections calibrated to official Swiss energy policy targets, enabling a reproducible and policy-relevant analysis. We address three sequential research subquestions, each building on the previous: (i) where are the binding network constraints that limit the FFOR, and how does the FFOR vary across seasons and DER types; (ii) given those binding constraints, which BESS placement maximises the aggregated flexibility at the PCC; and (iii) starting from the optimal spatial configuration, how much additional flexibility can be unlocked by relaxing voltage bounds, increasing DER penetration to 2040 and 2050 levels, and — as a methodological validation — replacing the linearised OPF with a full AC-OPF?

Overall, we want to answer the following question:

How do network constraints, BESS placement, and future DER penetration jointly determine the aggregated flexibility potential of a Swiss MV distribution grid, and how accurate is the linearised OPF approximation in capturing these effects?

Methodology Following the approach of [1], we quantify aggregated distribution grid flexibility through the computation of the Feasible Flexibility Operating Region (FFOR) at the PCC. The FFOR is approximated as a convex polytope in the active–reactive power (PQ) plane, obtained by iteratively solving a linearised Optimal Power Flow (OPF) problem along multiple directions [2]. The OPF is formulated using a first-order Taylor expansion of the AC power flow equations and explicitly enforces line current limits and nodal voltage limits ($U \in [0.95,\, 1.05]$ p.u.). Device-level models follow the same formulations as in the reference article: BESS are modelled with an apparent-power polygon and inter-temporal State-of-Charge constraints; PV systems with a curtailable active power and a power-factor range; and heat pumps as controllable loads with first-order thermal building dynamics and comfort temperature bounds.

The baseline operation at each timestep is derived directly from the Zapparoli & Oneto dataset [4]: non-controllable load profiles scaled by nodal peak demand, PV generation profiles, and HP consumption computed from the RC thermal model with local temperature data. This approach supersedes the statistical inference of device capacities used in [1], providing a reproducible and data-driven baseline.

The three analyses are structured as a sequential workstream:

- **Analysis 1 – Baseline FFOR, Binding Constraints, and Seasonal Variation** (Student 1): This workstream establishes the foundational results on which Analyses 2 and 3 depend. For the 2030 DER scenario, Student 1 computes: (a) the single-timestep and multi-timestep FFOR for two representative snapshots (sunny summer afternoon and cold winter evening), with sustained durations from 15 min to 8 h; (b) the flexibility gain from including heat pumps and PV as active flexibility providers, showing the incremental contribution of each DER type; (c) the seasonal variation of maximum active power flexibility over a four-month window (September–December); and (d) a systematic identification of the *binding constraints* — which specific lines and voltage nodes become active in each direction of the PQ plane, and at which timesteps. This constraint map is the direct input to Analysis 2, guiding the selection of candidate nodes for BESS placement.
- **Analysis 2 – BESS Location Sensitivity** (Student 2): Building on the binding-constraint map produced by Student 1, Student 2 evaluates how the placement of the community-scale BESS (from the 2030 dataset) affects the aggregated flexibility at the PCC. Rather than sweeping all nodes blindly, the candidate locations are selected based on the constraint topology identified in Analysis 1: nodes upstream of bottleneck lines, nodes at feeder midpoints, and nodes at feeder endpoints. The FFOR volume is computed for each placement and mapped as a function of electrical distance from the PCC. Student 2 identifies the optimal placement and quantifies the physical mechanism (line saturation vs. voltage constraint) that makes certain locations more effective than others. The optimal-placement scenario is then passed to Student 3 as the reference configuration for boundary-expansion experiments.
- **Analysis 3 – Physical Boundary Expansion and Future Scenarios** (Student 3): Starting from the optimal spatial configuration identified by Student 2, Student 3 quantifies how much additional flexibility can be unlocked by relaxing physical constraints and by increasing DER penetration. Three sub-analyses are performed. First, *voltage constraint relaxation*: the FFOR is recomputed with wider voltage bounds ($U \in [0.93,\, 1.07]$ p.u.) and the percentage gain in FFOR volume is reported as a sensitivity measure for DSO operational policies. Second, *future scenario analysis*: the full pipeline is repeated with the 2040 and 2050 DER allocations from [4], tracking whether aggregated flexibility scales monotonically or exhibits saturation effects due to feeder overloading — extending the analysis of Fig. 9 in [1] with data-driven national projections. Third, *linearisation accuracy*: on two to three representative snapshots, the FFOR obtained with the linearised power flow is compared against the FFOR computed

with the full AC-OPF (DistFlow formulation) available in `pandapower`. This provides an explicit validation of the linearisation error and directly addresses the open question raised in [1] regarding the accuracy gap between linear and nonlinear OPF approaches, without requiring the full pipeline to run under AC-OPF.

The complete optimisation model is implemented in Python using `pandapower` for grid modelling and power flow, and `scipy.optimize` (LP solver) for the linearised OPF iterations. The AC-OPF comparison in Analysis 3 uses `pandapower`'s built-in `runopp()` solver. All data, code, and results will be made available as a commented and structured zip archive.

References

[1] Jan Brändle, Julie Rousseau, Pulkit Nahata, and Gabriela Hug. On the flexibility potential of a Swiss distribution grid: Opportunities and limitations. *arXiv preprint arXiv:2510.13449*, 2025. Power Systems Laboratory, ETH Zürich.

[2] Luis Lopez, Alejandro J. Gonzalez-Castellanos, David Pozo, Mardavij Roozbehani, and Munther A. Dahleh. QuickFlex: A fast algorithm for flexible region construction for the TSO-DSO coordination. In *International Conference on Smart Energy Systems and Technologies (SEST)*, 2021. doi: 10.1109/SEST50973.2021.9543349.

[3] Alfredo Ernesto Oneto, Filippo Tettamanti, Blazhe Gjorgiev, and Giovanni Sansavini. Synthetic low- and medium-voltage grids for Switzerland, 2025. Version v1.0, CC-BY 4.0.

[4] Lorenzo Zapparoli, Alfredo Ernesto Oneto, María Parajeles Herrera, Blazhe Gjorgiev, Gabriela Hug, and Giovanni Sansavini. Future deployment and flexibility of distributed energy resources in the distribution grids of Switzerland, 2025. Version v3, CC-BY 4.0.